Band-1 Hadamard Order 6388 — Miyamoto (Paley-II H_1596 + C_1598 over prime 1597); Miyamoto-only Bonus (NOT in SPK); Gap 2052 Still OPEN

H_1596/C_1598/H_6388 Miyamoto PASS; Paley-I/Double SKIP; Paley-I/II on 6388 SKIP; 6388 Miyamoto-only (NOT in SPK); gap 2052 still OPEN; conjecture OPEN; GATE-04 LOCKED

C.M. Baird; ZoraASI mission-control Band-1 dash

2026-09-15

Document class. Scientific computational / construction certificate (Band-1).
Program. Open-mathematics collapse list · Human+AI attack order · ZoraASI mission-control.
Parents. `2026-09-15_Band1_Hadamard_6212_Miyamoto`; `2026-09-15_Band1_Hadamard_6196_Miyamoto`.
Lab. `/workspace/experiments/band1/hadamard_6388_miyamoto/`.
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Honesty stamp (non-negotiable).
This memo claims **finite constructions of orders 1596 and 6388 only** (verified $\max|HH^\top - nI| = 0$), via Miyamoto with preferred Paley-II seed H_{1596} = Paley-II($q = 797$) and symmetric Paley conference C_{1598} over prime $q = 1597 \equiv 1 \pmod 4$. Paley-I for the seed ($q = 1595 = 5 \cdot 11 \cdot 29$) **FAIL**s; Double half=798 **FAIL**s ($q = 797 \equiv 1 \pmod 4$). Classical SPK paths for order 6388 **FAIL**: Paley-I ($q = 6387 = 3 \cdot 2129$ not prime); Paley-II ($q = 3193 = 31 \cdot 103$ not prime); Kronecker (4×1597). Therefore order **6388 is a Miyamoto-only bonus (NOT in SPK)** (same honesty pattern as 6196/5956/5812/5476). It **does not** claim the full Hadamard conjecture. It **does not** stamp Clay-closed or finished-ToE. GATE-04 collection remains **LOCKED**. No RH / P≠NP / YM / Frankl / NS-solved / AC-solved / Hadamard-conjecture-solved claims. Standard constructive gap **2052** remains **OPEN** under Sylvester∪Paley∪Kronecker (SPK) closure. `clay_closed = false`.

Abstract

Finite constructions of orders **1596** and **6388** via Miyamoto.

- H_{1596} = Paley-II ($q = 797$) — preferred; Paley-I / Double SKIP
- C_{1598} = Paley conference over prime 1597
- H_{6388} = Miyamoto(H_{1596}, C_{1598})
- Paley-I / Paley-II for 6388: **SKIP**

SPK: 6388 Miyamoto-only bonus (NOT in SPK). Seed 1596 enters SPK via Paley-II. Prior SPK seeds 388/1552/6212 + 1548 retained in ONBOX.
Gap 2052: OPEN. **Conjecture:** OPEN. **clay_closed:** false. **GATE-04:** LOCKED.

`python3 verify_hadamard_6388.py`

Results. overall_pass=true; units 80/80; wall 215.8041s; all max|HH^T-nI|=0.

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Arithmetic

- Target $n = 6388 = 4 \cdot 1597$; 1597 prime $\equiv 1 \pmod 4$ → Miyamoto-eligible.
- Need H_{1596} and C_{1598} (Paley conference over prime 1597).
- Seed preferred: Paley-II $q = 797$ (prime $\equiv 1 \pmod 4$); $2(q+1) = 1596$.
- Paley-I seed $q = 1595$ **FAIL** ($5 \cdot 11 \cdot 29$); Double half=798 **FAIL**.
- SPK: Paley-I $q = 6387$ **FAIL**; Paley-II $q = 3193$ **FAIL**; Kronecker **FAIL** → **Miyamoto-only**.

Constructions (verified)

Object	Method	max|HH^T-nI|	Wall (s)
H_1596 Paley-II	Paley-II q=797 (preferred)	0	2.33
H_1596 Paley-I	SKIP q=1595=5 · 11 · 29 not prime	—	—
H_1596 Double	SKIP q=797 1 mod 4	—	—
C_1598	Paley conf q=1597	0	3.244
H_6388 Miyamoto	Miyamoto(H_1596, C_1598)	0	207.647
H_6388 Paley-I	SKIP q=6387=3 · 2129 not prime	—	—
H_6388 Paley-II	SKIP q=3193=31 · 103 not prime	—	—

SPK closure note

Seeds include prior SPK ships through **6212** lineage (**388**/**1552**/**6212** from that ship; **1548** from 6196 ship; **NOT** 6196/6388) + verified **1596** (Paley-II); **6388 NOT folded** (Miyamoto-only);

exclude Miyamoto-only bonuses including **6388/6196/5956/5812/5476**. Known count 1314. Next constructive gap **2052** (OPEN).

Quantity	Value
6388 in SPK	**False** (Miyamoto-only bonus)
1596 in SPK	**True** (Paley-II)
6212 in SPK	**True** (prior Paley-I)
6196 in SPK	**False** (prior Miyamoto-only)
Next constructive gap	**2052** (OPEN)

Next Miyamoto targets (after q=1597)

q	n=4q	need H	build notes
1601	6404	1600	['Kronecker(4x400)']
1609	6436	1608	['PaleyI(q=1607)']
1613	6452	1612	[]
1621	6484	1620	['PaleyI(q=1619)', 'PaleyII(q=809)']
1637	6548	1636	[]
1657	6628	1656	['Double(PaleyI half=828)']

Honesty

clay_closed = false · GATE-04 LOCKED · Hadamard conjecture OPEN · gap 2052 OPEN · no RH/P!=NP/YM/Frankl/NS/AC/Clay-closed claims.